Achievements Screen Spec

Oct 19, 2026 · @Lefika Thobo Moalosi

Overview

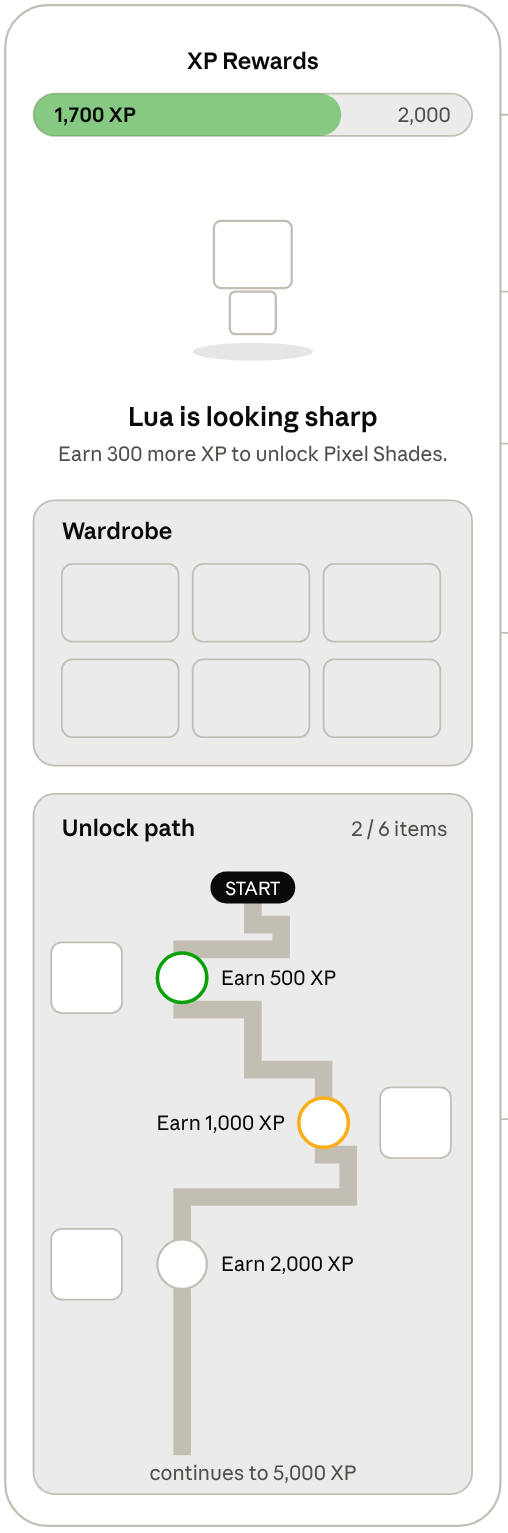
The achievements screen turns XP into rewards for the player's pixel avatar, Lua: 17 rewards unlock on a path from 500 to 15,000 XP, made up of all 13 items of clothing and 4 colour skins. Players see Lua, their XP and the next reward, a wardrobe of clothes and skins, and a staircase path with an animated preview of each reward. A gold Shop button opens a sheet that sells 16 costume skins for real money.

The reference build is the **Lua XP Rewards** prototype (open it). It is private until Lefika shares it. Where this spec and the prototype disagree, ask before building.

Screen layout

The screen is one scrolling column on every device, in the app's light style: a title row with the Shop button, the XP bar, avatar, status line, the wardrobe card (clothes, then skins), then the unlock path card. On larger screens it stays phone width (about 430 px), centred, like the app's other screens. The shop opens as a bottom sheet over this column, never as a separate screen.

On every device the screen is one phone-width column



XP bar
Total inside, next milestone at the right

Avatar
On white; idle motion, toast, tap to hop

Status line
Mood headline, XP to the next item

Wardrobe
6 items in 3 columns
Tap to wear or take off

Unlock path
Staircase pixel road, START to 5,000 XP
Badge on the road, label on the inner side
Animated preview on the outer side

achievements screen on a phone · 4 regions, top to bottom

The background is white with 16 px side margins. The wardrobe and the path sit in light grey cards with 18 px corners, like the stats card on the Health Bar screen.

Avatar stage

The avatar sits large on the white background with no card behind it, the way Lua appears on the Health Bar screen, wearing everything currently equipped. The stage area is full column width and slightly shorter than it is wide (height about 0.92 of the width).

- **Avatar:** the 12×16 pixel sprite, centred, feet on a soft shadow ellipse about 88% of the way down. Always scale with nearest-neighbour filtering, never smoothed.
- **Idle loop:** the head dips gently (about a quarter of an art pixel, 2.6 s cycle), the hands sway the opposite way, and the eyes blink for 0.13 s every 2.2 to 5.4 s. Every few seconds the eyes glide one pixel left or right and back.
- **Tap:** a small hop with happy eyes, about 0.85 s: squash down, rise, land with a squash, settle. Ignore taps while a hop is playing.
- **Toast:** a dark pill (#1d1b24, white text) at the top of the stage with the item icon and text such as "Unlocked Pixel Shades!". It slides in and hides after 2.6 s.
- **Sneakers and Rain Boots:** the soles add 2 art pixels under the feet, so the whole avatar rises 2 art pixels while either is worn (eased, not a jump).
- **Colour skins:** Classic, Mint, Strawberry, Midnight or Gold recolour Lua's own pixels (outline, white, lavender, purple). Clothes keep their colours and are worn on top.
- **Costume skins:** a costume (Ninja, Samurai and so on) is a complete outfit. While one is on, the worn clothes are hidden but stay equipped, and the avatar does not lift for shoes. Putting on any item of clothing takes the costume off and returns Lua to the last colour skin used.
- **Changing skin:** the avatar hops, sparkles burst and the toast reads "Mint skin on!"; the new look pops in (scale 1.35 to 1 over 0.35 s).

Title row, XP bar and status line

The top row holds the centred title "Achievements" with the gold Shop button on the right (see Shop). The XP bar sits under it, styled like the health bar. A status line under the avatar says how Lua is doing and what comes next.

- **XP bar:** a 30 px tall, fully rounded pill with a light grey track (#e6e6ea). It fills with a green-to-yellow gradient (#22a24a to #e8c21b) for progress from the previous milestone to the next, so at 1,700 XP it is 40% full (1,500 to 2,000).
- **Inside the bar:** the total on the left ("1,700 XP", thousands separated) and the next milestone on the right in grey ("2,000"). When XP is earned, the total counts up over 1.1 s, easing out.
- **Status headline:** large and bold, centred: "Lua is just getting started" with no rewards, "Lua is looking sharp" with some, "Lua has every reward!" with all 17.
- **Status line:** grey, centred: "Earn 300 more XP to unlock **Ball Cap**." After the last reward: "Every reward on the path is unlocked."

Wardrobe

The wardrobe is a light grey card (#f3f3f5) with two 3-column grids. "Wardrobe" (hint on the right: "Tap an item to wear it") lists all 13 items of clothing in path order. Under it, "Skins" (hint: "Tap a skin to use it") lists the 5 colour skins, then any costumes bought in the shop. Each tile is white (1.5 px border #e3e2e9, 12 px corners) with the icon, its name and a one-line status.

State	Tile look	Status text	On tap
Locked (clothes or colour skin)	Icon grey and darkened	Locked · 2,000 XP	Toast: "Earn 2,000 XP to unlock"
Owned item	Normal	Tap to wear	Puts the item on
Wearing	Green border (#22a24a), pale green tile (#eef8f1)	Wearing (green)	Takes the item off
Equipped under a costume	Normal	Under costume	Takes the costume off, so the item shows again
Owned skin	Normal	Tap to use	Switches to that skin
Skin in use	Green border, pale green tile	Using (green)	Nothing

- **One item per slot.** Putting on an item replaces whatever is in its slot. Head: Cozy Beanie, Ball Cap, Headphones, Wizard Hat, Gold Crown. Legs: Denim Pants, Track Shorts. Feet: Sneakers, Rain Boots. Top: Hoodie. Back: Hero Cape. Neck: Striped Scarf.

Face: Pixel Shades.

- **Hoodie:** its hood covers the head all round and its sleeves ride on the arms; hats, headphones and shades sit on top of the hood.
- **Put on:** the item pops onto the avatar, scaling from 1.35 to 1 over 0.35 s. If a costume is on, it comes off first.
- **Take off:** the item floats up about 1.6 art pixels and fades out over 0.35 s.

Unlock path

The path sits in its own light grey card below the wardrobe. It is a pixel road that runs from a dark START pill at the top down through 17 milestones in a staircase: straight vertical and horizontal runs with square corners, never curves or diagonals. The card is headed "Unlock path", with a count such as "2 / 17 rewards" on the right.

Road shape

- Build it on a tile grid: 80 columns across the path area. Milestones alternate between columns 30 and 50, 28 rows apart; START is at column 40.
- Each stretch to the next milestone goes down 8 rows, sideways to a detour column, down 10 rows, sideways to the next milestone's column, then down to it. Detour columns in order: 48, 40, 54, 38, 44, 26, 56, 36, 46, 24, 52, 40, 58, 34, 44, 26, 50.
- The road is 6 tiles wide with a 1-tile dark border (#4d485c, the sprite outline colour) all round, square at the corners. Unearned road is a light grey checker (#e9e7ef / #dedbe7) in 2×2-tile squares.

Progress fill

- Tiles turn green in route order as XP rises. Between two milestones the fill is proportional: 1,700 XP fills 70% of the 1,000 to 2,000 stretch.
- Filled road is a green checker (#22a24a / #30b257). The 2 tiles at the front edge are light green (#9be8b3).
- Each newly filled tile flashes white (#ffffff) for 140 ms, then bright green until 420 ms, then settles. Tiles already earned when the screen opens do not flash.

Milestones

- **Badge:** a 56 px white circle on the road at each milestone, showing the item icon. Done = green ring and a green check. Next = gold ring (#e2b324) that pulses every 1.8 s. Locked = pale grey circle (#ecebf1), faded grey icon, small lock.

- **Label:** on the side of the badge facing the middle of the screen: "Earn 1,000 XP" in the pixel font, the item name below it (green once unlocked), on a backing in the card colour so it stays readable over the road.
- **Preview:** on the outer side of each milestone, alternating left and right, a white rounded tile (1.5 px border #e3e2e9, 14 px corners) playing the avatar wearing only that item, in the colour skin in use (the last colour skin if a costume is on). A skin reward shows Lua in that skin. A tag under it reads "Preview", or "Unlocked" on green once earned.

Reward	Preview pose
Sneakers	Quick happy hops with a squash on each landing
Denim Pants	Side-to-side strut, eyes glancing the way it leans
Ball Cap	Looks one way, then the other, with a little nod
Pixel Shades	Cool nod, then a white shine slides across the lenses
Mint, Strawberry, Midnight and Gold Skins	Happy hop in the new colours with twinkles
Cozy Beanie	Shivers in falling snow, then a happy, cosy face
Track Shorts	Jogging on the spot, arms pumping
Striped Scarf	Eyes closed, leaning into gusts of wind, scarf flapping
Headphones	Eyes closed, nodding to the beat, purple music notes drifting up
Rain Boots	Stomping hops in falling blue rain
Hoodie	Cosy sway, eyes closed then happy
Hero Cape	Heroic leap with the wind behind
Wizard Hat	Happy sway with twinkles round the hat
Gold Crown	Proud bounce with gold and orange twinkles round the crown

Unlock flow and rules

A reward unlocks the moment the counting-up XP number passes its threshold: clothing is put on automatically, and a colour skin is applied. Gaps between rewards grow from 500 to 1,500 XP, so later rewards take longer.

XP	Reward	Slot
500	Sneakers	Feet
1,000	Denim Pants	Legs
1,500	Ball Cap	Head
2,000	Pixel Shades	Face
2,500	Mint Skin	Colour skin
3,000	Cozy Beanie	Head
4,000	Track Shorts	Legs
5,000	Striped Scarf	Neck
6,000	Headphones	Head
7,000	Strawberry Skin	Colour skin
8,000	Rain Boots	Feet
9,000	Hoodie	Top
10,000	Midnight Skin	Colour skin
11,500	Hero Cape	Back
13,000	Wizard Hat	Head
14,000	Gold Skin	Colour skin
15,000	Gold Crown	Head

What happens when XP is earned:

1. The XP number counts up over 1.1 s, and the road fills tile by tile in step with it.
2. When the count passes a clothing threshold, the item unlocks:
    a. If a costume is on, it comes off and Lua returns to the last colour skin.

   b. The avatar hops with happy eyes.
   c. The item drops onto the avatar from about 9 art pixels above with a bounce (0.9 s). Anything already in that slot fades out.
   d. Pixel sparkles (purple, gold, light blue, green) burst from the item as it lands.
   e. The toast shows "Unlocked <item>!".
   f. The badge turns done, the next badge gets the gold ring, the wardrobe tile shows Wearing, and the preview tag switches to Unlocked.
3. When the count passes a skin threshold, Lua hops, the skin is applied with sparkles, the toast shows "Unlocked Mint Skin!", and the skin's tile in the wardrobe shows Using.
4. If one gain crosses several thresholds, the unlocks play in order, 1 s apart.

Shop

The Shop sells 16 costume skins for real money; clothes and colour skins are only earned on the path. It opens from the Shop button as a bottom sheet over the screen, with no separate screen.

Shop button: a gold pill at the right of the title row, 38 px tall with 10 px corners, a gradient from #ffe27a through #f7c234 (55%) to #e9a514, and a 2 px #4d485c border with a hard shadow 3 px straight down. The face presses into the shadow when tapped. It holds a white shopping-bag icon and "SHOP" in #3a2a06. A white shine (14 px, skewed −20°) sweeps across every 4.5 s, and a red NEW tag (#e0524a) bobs on the corner until the shop is first opened.

Sheet: slides up over a dimmed backdrop, with a grabber, the title "Shop" and a close button. A tap on the backdrop or Escape closes it. One card, "Costume skins" (hint: "Full outfits for Lua"), holds a 3-column grid of tiles: a still front view of the costume, its name and a button.

Costume	Price	Costume	Price
Soldier	$1.99	Pirate	$1.99
Samurai	$2.99	Chef	$0.99
Office Man	$0.99	Firefighter	$1.99
Ninja	$2.99	Doctor	$1.49
Spy	$2.49	Cowboy	$1.99

Costume	Price	Costume	Price
Student	$0.99	Robot	$2.49
Astronaut	$2.99	Viking	$2.49
Knight	$2.49	Superhero	$2.99

Prices are placeholders. The tile button shows the price in green when not owned, "Use" when owned, and "Using" when it is on.

1. Tapping a price opens a confirm step inside the sheet: the costume, "Buy Ninja skin?", and Cancel / "Buy $2.99". In the app, this step hands off to the App Store or Google Play checkout.
2. After the purchase the sheet closes, the costume goes on with a hop, sparkles and the toast "Ninja skin on!", and it appears in the wardrobe's Skins row.
3. "Use" puts an owned costume back on and closes the sheet.

Visual style

The screen uses the app's light theme, matching the Health Bar screen: white background, light grey rounded cards, green for progress and the main button, and one pixel font throughout.

Use	Colour
Screen background, item tiles, previews, badges	#ffffff
Cards (wardrobe, unlock path, shop)	#f3f3f5, 18 px radius, faint shadow
Tile and card borders	#e3e2e9
Main text, START pill, toast	#1d1b24
Secondary text	#6c6879
Green (progress, wearing, unlocked, main button)	#22a24a, pressed #1b8a3e
XP bar gradient	#22a24a to #e8c21b on a #e6e6ea track
Next-milestone gold	#e2b324

- **Sprite colours (Classic):** outline #4d485c, white #eeebf8, lavender #cfcbdd, purple #ac9dc7. Clothing and costume colours are baked into the sprite sheets.
- **Colour skins** swap those four colours. Mint: white #effcf5, lavender #c4ead6, purple #7fcda6. Strawberry: #fff0f3, #f7c6d2, #e98aa3. Midnight: outline #1f1b33, #8f88b8, #6a6394, #433d6b. Gold: outline #5e4416, #fff6d6, #f3d98a, #d9a93c.
- **Effects on white:** sparkles use purple, gold, blue and green; snow is a soft blue-grey (#a9b8dc); rain is blue (#5f96dc); music notes are purple (#6b4fc9); twinkles are gold and orange. Plain white effects would vanish on the white background.

Assets

All art comes as 63 sheets (2048×2048 PNG) in `lua_sheets_v3.zip`, each a 4×4 grid of 512 px cells with one animation per row. Every cell is a 32×32 grid of art pixels at 16 px each, and the avatar's feet sit on y = 464 in every cell of every sheet. `sprites.json` in the zip lists every sheet, its rows and cells, the path order with XP, and the skins and prices. `make_sheets.py` regenerates everything. `luaSprites.ts` and `road.ts` from the earlier kit still describe the 6-item path and need updating before use.

Sheets	Count	Contents
`avatar_body.png`, `avatar_body_mint.png`, `_strawberry`, `_midnight`, `_gold`	5	Lua in each colour skin. Rows: idle, blink & glance, hop, happy
`item_<id>.png`	13	One per item of clothing. Same rows and frames as the body; stackable layers that work on every colour skin
`costume_<id>.png`	16	One per costume: the whole avatar in that costume, same rows. Replaces the body and all clothes while worn
`preview_poses_1.png` to `preview_poses_5.png`, plus a set per colour skin	25	The 17 path previews in path order, 4 per sheet; the unlock sparkle burst is sheet 5, row 1
`item_icons.png`, `item_icons_locked.png`	2	Clothing icons, unlocked and greyed

Sheets	Count	Contents
`skin_icons.png`	1	Colour skin tiles: Classic, Mint, Strawberry, Midnight in row 0, Gold at row 2; locked versions in row 1
`skins_costumes.png`	1	A still front view of each costume, for shop tiles

- **Dressing the avatar:** stack the body sheet for the colour skin and the worn item sheets inside one animated view, bottom to top: body, back (cape), legs, top (hoodie), feet, neck, face, head. Each item sheet holds only the pixels that item changes, so the layers line up with no offsets.
- **Costume on:** draw only that costume's sheet, with no body or clothes underneath.
- **Shoe lift:** while Sneakers or Rain Boots are worn (and no costume is on), raise the whole stack by 32 cell px, scaled to the on-screen size.
- **Previews:** use the preview sheets for the colour skin in use, or for the last colour skin if a costume is on.
- **Road:** not in the sheets. Draw it in code from the tile grid in the Unlock path section.

React Native implementation notes

Run every animation on the UI thread with Reanimated, the same way `HealthStatusSprite` steps frames with a transform; never step frames with React state.

- **Frame stepping:** reuse the `HealthStatusSprite` pattern (`withRepeat` + `withTiming` to 4, `Math.floor` , `translateX` / `translateY` on the sheet). Play the 4-frame rows at about 6 to 8 fps.
- **Smooth motion on top:** the sheets only have 4 frames per row. Add the bob, squash and stretch, item drop bounce, pop and fade with Reanimated transforms on the stacked view.
- **Crisp pixels:** draw the sheets with nearest-neighbour filtering. React Native Skia (`FilterMode.Nearest`) does this. Plain `Image` will blur unless the art is shown at an exact multiple of its size.
- **Memory:** each sheet uses about 16 MB once decoded, so a fully dressed avatar (up to 8 layers) is around 130 MB; a costume is one sheet. If that is too much, ask for quarter-size sheets (512×512) and scale them up with Skia.

- **Purchases:** costume skins are digital goods, so Apple and Google require in-app purchases for them. The prototype only simulates the confirm step.
- **Road:** compute the tile list once from the route, then draw it with Skia rects. Only redraw while the fill is moving or tiles are flashing.
- **Reduce Motion:** when `AccessibilityInfo.isReduceMotionEnabled()` is true, turn off the idle bob, hops, bounces, sparkles, tile flashes and the badge pulse.
- **Accessibility:** give the avatar, each wardrobe card and each milestone a label, for example "Pixel Shades, locked, unlocks at 2,000 XP".

Prototype-only parts and open questions

The four demo buttons in the prototype (+150 XP, +1,000 XP, Max out, Reset) only simulate XP and are not part of the app, and neither is its starting state of 1,200 XP. Purchases in the prototype are simulated and take no payment.

- [ ] Where does XP come from, and when should the screen hear about new XP (live, or on next open)?
- [ ] Where are unlocked and equipped items stored, and do they sync across devices?
- [ ] What happens past 15,000 XP: more milestones later, or the path ends?
- [ ] Should unlocking always put the item on, or ask first?
- [ ] Should tapping a preview or badge do anything, such as open the item?
- [ ] Does the avatar also appear on other screens wearing the same outfit?
- [ ] Final product IDs and prices for the 16 costumes (the $0.99 to $2.99 prices are placeholders), and how restoring purchases works.
- [ ] Should a costume worn when new clothing unlocks come off automatically (as the prototype does), or stay on?